



2020-05-07 Performance tests analysis

Mondo Trgovina - Saso Mange

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DOCUMENT REVISION HISTORY

Date	Revision	Author	Summary of Changes
07/05/2020	1.0	franciszek.bieg@sap.com	Final

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INTRODUCTION

Report presents analysis of the Saso Mange performance problems on P1 environment. Traffic has been generated by Customer scripts. Report contains only analysis of the system after executed tests.

RECOMMENDATIONS

Recommendations SOLR

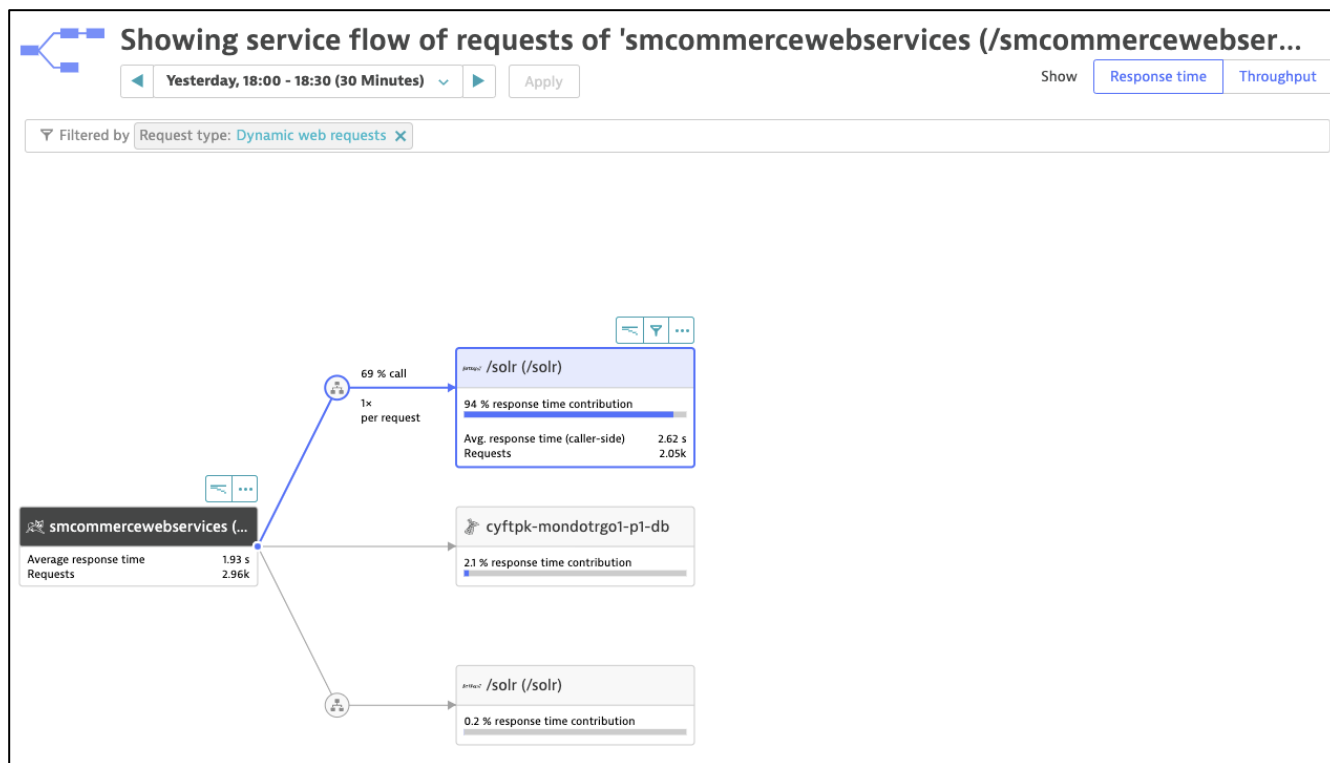
- Increase number of replicas.
- Investigate if all search requests from frontend application really need facet values.
- Try to limit amount of facet attributes if possible from business perspective.
- Limit amount of facets for general search.
- Limit amount of facets for /smcommercewebservices/v2/smrs/catalogs/categories
- Conduct end-to-end performance tests with scenario real customer path to properly scale infrastructure

Recommendation WISHLIST

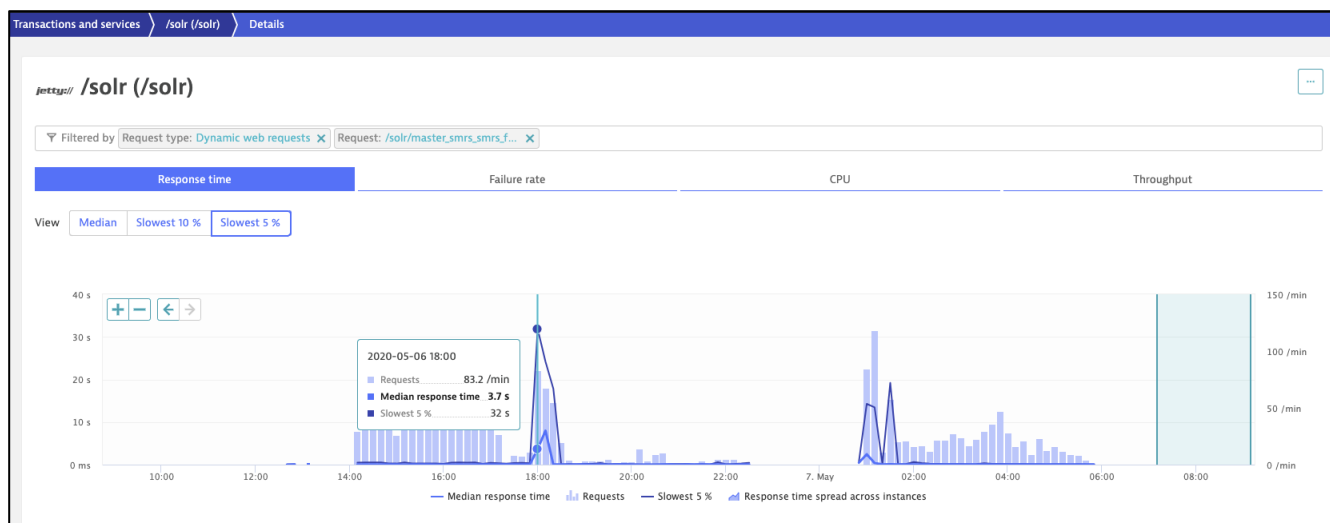
Validate list of populators executed by SmWishlistFacadeImpl.getWishListProductsAndCategories method. Try to remove or refactor populators populatorsProductStockPopulator, ProductClassificationPopulator, SmCustomCategoryHierarchyPopulator

PROBLEM WITH SOLR LONG RESPONSES

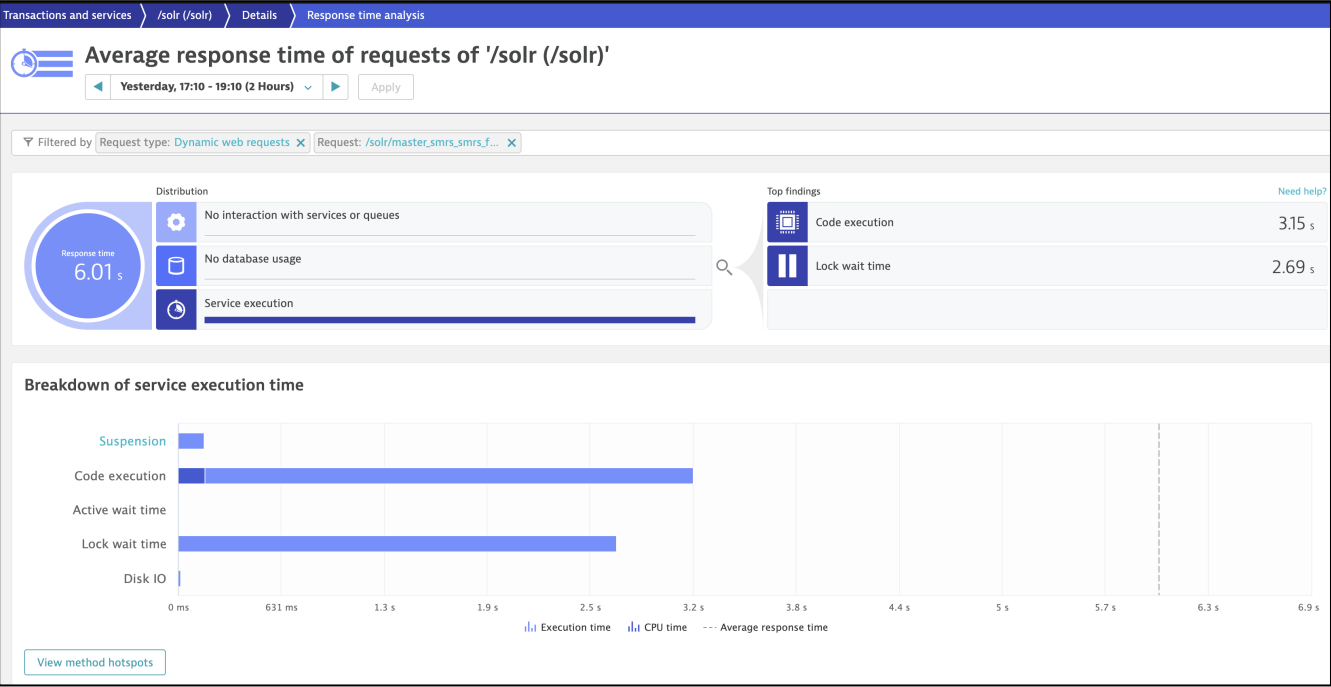
Influence of SOLR longer response times:



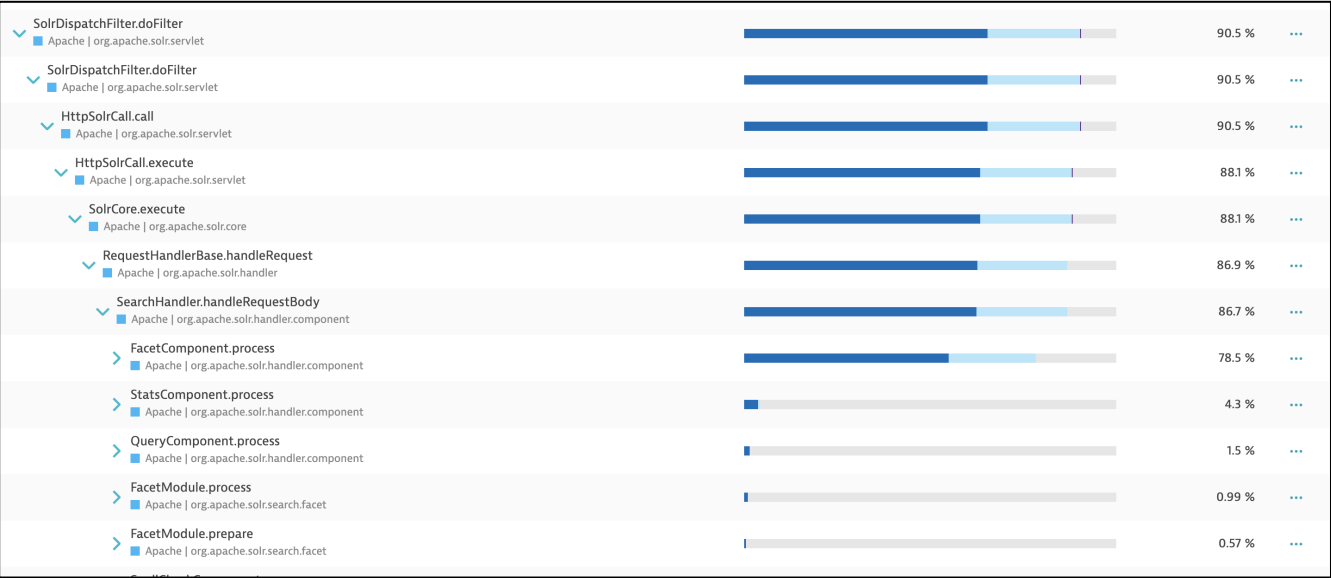
Longer response times on SOLR node occurred at 18:00 2020-05-07:



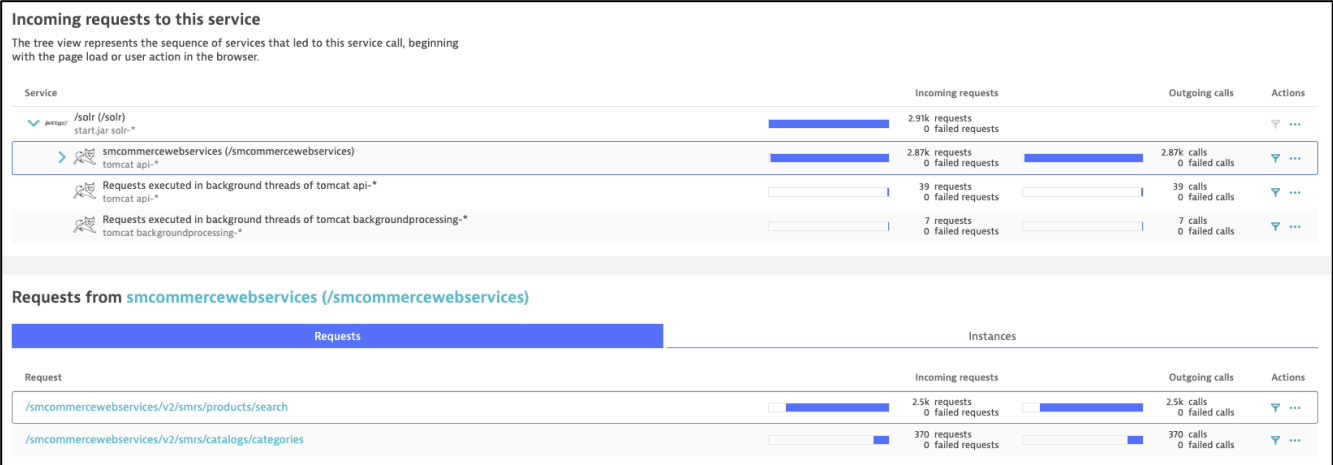
Breakdown of service execution time for SOLR node:



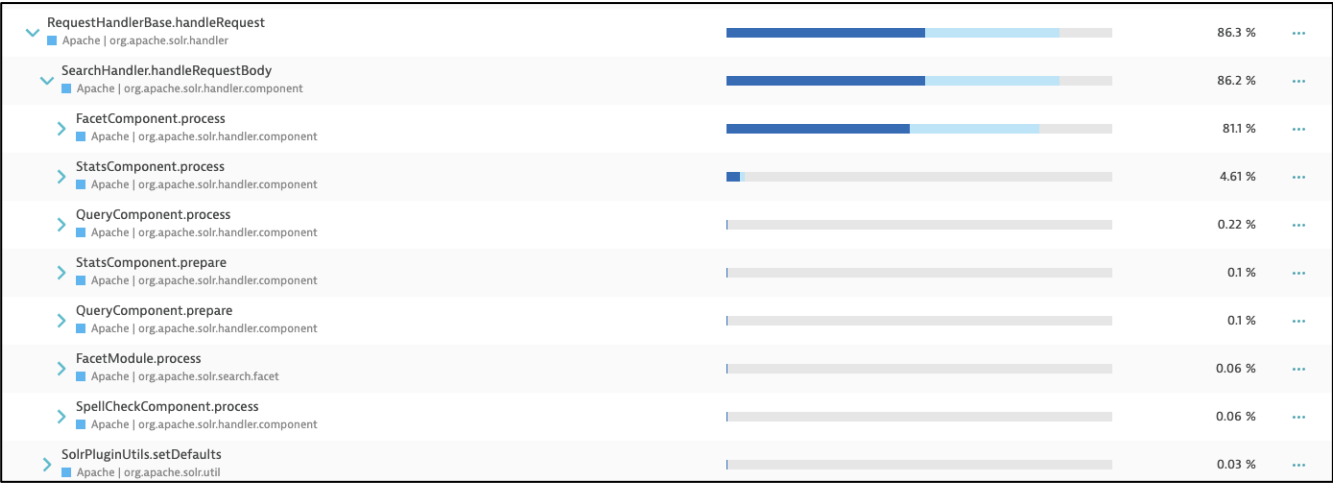
Method hotspot shows long processing time of facets:



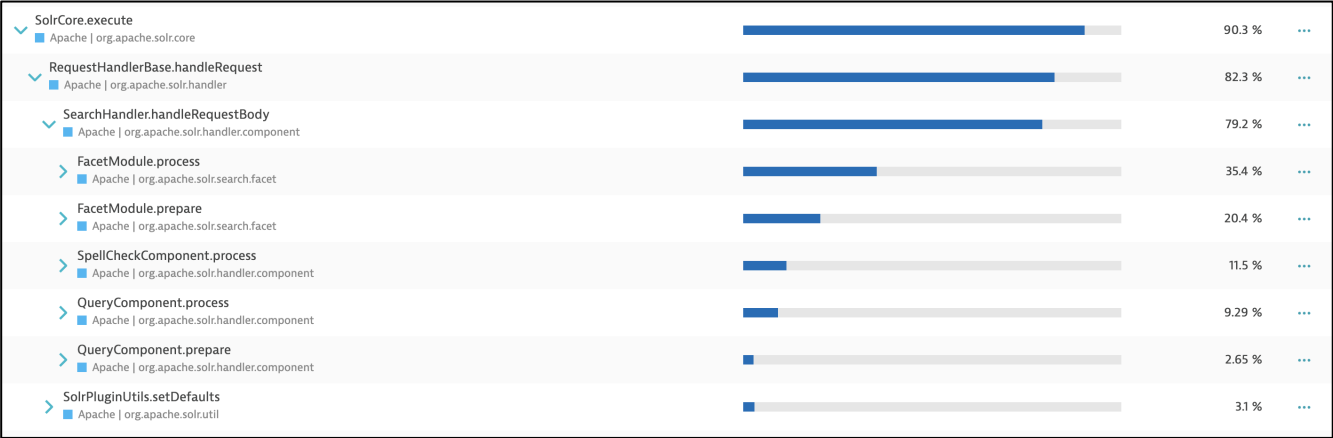
There are two endpoints which send requests to SOLR:



For requests related to **/smcommercewebservices/v2/smrts/products/search** endpoint, method hot spot shows that SOLR response time is **affected in 81% by facets processing** which is have CPU operation for SOLR:



For requests related to **/smcommercewebservices/v2/smrscatalogs/categories** endpoint, method hot spot shows that SOLR response time is affected in **55,8%** by facets processing which is have CPU operation for SOLR:



Each query from **/catalogs/categories** or **/products/search** is querying 1106 facet fields:

```
2020-05-07 12:33:09.444 INFO (qtp42102625-3034) [c:master_smrs_smrs_flip s:shard1 r:core_node4
x:master_smrs_smrs_flip_shard1_replica_n3] o.a.s.c.S.Request [master_smrs_smrs_flip_shard1_replica_n3]
webapp=/solr path=/select params={facet.field=proizvodjac-displayport-kablovi-kablovi_string_mv &
```

```
facet.field=stanje-ostali-elektro-delovi_string_mv &
facet.field=tip-hrane-za-ptice-i-male-zivotinje_string_mv &
```

```
facet.field=ostecenje-kombajni_string_mv &
```

```
facet.field=stanje-rucni-alat_string_mv &
```

```
facet.field=bazen-planina-ponuda-privatan-smestaj_string_mv &
```

```
facet.field=rooms_heating_type_string_mv &
```

```
facet.field=proizvodjac-hdmi-kablovi-kablovi_string_mv &
```

```
facet.field=stanje-gradjevinski-materijali_string_mv &
```

```
facet.field=pol-deteta-namestaj-za-decu_string_mv &
```

```
facet.field=proizvodjac-retro-konzole_string_mv &
```

```
facet.field=stanje-medicinska-oprema-i-pomagala_string_mv &
```

```
facet.field=broj-sedista-autobusi_string_mv &
```

```
facet.field=dizajn_string_mv &
```

```
facet.field=stanje-analogni-audio-video-adapteri-i-kablovi-kablovi-i-adapteri_string_mv &
```

```
facet.field=proizvodjac-mikrotalasne-rerne_string_mv &
```

```
facet.field=tip-softvera_string_mv &
```

```
facet.field=konekcija-misevi_string_mv &
```

```
facet.field=konektor-analogni-audio-video-adapteri-i-kablovi-kablovi_string_mv &
```

```
facet.field=menjac-autobusi_string_mv &
```

```
facet.field=dodatno-web-kamere_string_mv &
```

```
facet.field=materijal-ormari-i-garderoberi_string_mv &
```

```
facet.field=proizvodjac-vga-kablovi-kablovi_string_mv &
```

```
facet.field=broj-radnih-sati-traktori_string_mv &
```

```
facet.field=wi-fi-planina-ponuda-privatan-smestaj_string_mv &
```

```
facet.field=pet-frendli-city-break-ponuda-privatan-smestaj_string_mv &
```

```
facet.field=oprema-za-trzalacke-instrumente_string_mv &
```

```
facet.field=house_floor_sale_string_mv &
```

```

facet.field=tip-zimskog-sporta_string_mv &

facet.field=materijal-tapacirani-namestaj-garniture-antikvitetni-namestaj_string_mv &

facet.field=tip-masina-i-aparata_string_mv &

facet.field=konektor-vga-kablovi-kablovi-i-adapteri_string_mv &

facet.field=tip-masine-za-pranje-sudova_string_mv &

facet.field=slovni-raspored_string_mv &

facet.field=vrsta-goriva-kombajni_string_mv &

facet.field=materijal-kupatilski-namestaj-namestaj_string_mv &

facet.field=opticki-uredjaj_string_mv &

facet.field=proizvodjac-procesori_string_mv &

facet.field=luxury_filing_to_sale_string_mv &

facet.field=flat_heating_to_rent_string_mv &

facet.field=proizvodjac-sitna-bebi-oprema_string_mv &

facet.field=uzrast-sitna-bebi-oprema_string_mv &

facet.field=luxury_structure_sale_string_mv &

facet.field=proizvodjac-podloge-za-misa_string_mv &

facet.field=oblik-karoserije-svi-automobili_string_mv &

facet.field=proizvodjac-dodatna-oprema-za-a-v-uredjaje_string_mv &

facet.field=tip-prikolica-i-poluprikolica_string_mv &

facet.field=tip-male-gradjevine-masine_string_mv &

facet.field=proizvodjac-namestaj-za-decu_string_mv &

facet.field=tip-kozmetike_string_mv &

.

. cut

.

.

facet.field=stanje-vozila-agro-prikolice_string_mv &

```

```
facet.field=dodatno-a-v-plejeri_string_mv &

facet.field=vrsta-kazana_string_mv &

facet.field=plin-havarisani-automobili_string_mv &

facet.field=opema-za-harmonike_string_mv &

facet.field=sigurnost-kamioni-preko-7-5-t_string_mv &

facet.field=duzina-agro-prikolice_string_mv &

facet.field=stanje-kupatilski-namestaj-antikvitetni-namestaj_string_mv &

facet.field=tip-testera-za-vodu_string_mv &

facet.field=proizvodjac-hrana-za-pse-i-macke_string_mv &

facet.field=stanje-hi-fi-sistemi_string_mv &

facet.field=vesanje-kamioni-do-7-5-t_string_mv &

facet.field=flat_advertiser_to_rent_string_mv &

facet.field=house_bathrooms_to_sale_string_mv &

facet.field=tip-kreveta-kreveti_string_mv &

facet.field=diagonala-ekrana-monitori_string_mv &

facet.field=konektor-hdmi-kablovi-kablovi_string_mv &

facet.field=pet-frendli-more-first-minute_string_mv &

facet.field=proizvodjac-cd-plejeri_string_mv &

facet.field=tip-felni_string_mv &

facet.field=prevoz-more-ponuda-privatan-smestaj_string_mv &

facet.field=stanje-odeca-i-oprema_string_mv &

facet.field=pol-vga-kablovi-kablovi_string_mv &

facet.field=stanje-vozila-kombi-i-laka-dostavna-vozila_string_mv &

facet.field=proizvodjac-ostala-oprema-za-tv_string_mv &

facet.field=stanje-nauticka-oprema-i-delovi_string_mv &

_stateVer_=master_smrs_smrs_flip:46 &

fl=score,* &

start=0 &
```

```

fq=status_string:ACTIVE &

fq=vendor_string_mv:000001M2 &

sort=publishedTimestamp_long+desc,code_string+desc &

rows=100 &

version=2 &

q={!boost}{!lucene+v%3D$yq)} &

facet.limit=50 &

stats=true &

yq=*: * &

facet.mincount=1 &

facet=true &

wt=javabin &

facet.sort=count &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-kamioni-do-7-5-t_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_flat_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}predjeni-kilometri-svi-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_rent_double_mv &

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stats.field={!min%3Dtrue+max%3Dtrue}snaga-motora-havarisani-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-autobusi_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-kamperi-i-kamp-prikolice_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-motorcikli_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-kombajni_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_cottage_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_house_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-autobusi_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-prikljucne-masine_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godina-proizvodnje-havarisani-automobili_string_mv &

```

```

stats.field={!min%3Dtrue+max%3Dtrue}godiste-kombi-i-laka-dostavna-vozila_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_business_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-motorcikli_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-kamioni-preko-7-5-t_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_land_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}predjeni-kilometri-havarisani-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-cm3-svi-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_luxury_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}publishedTimestamp_long &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-motorcikli_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}snaga-motora-svi-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-agro-prikolice_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-kombajni_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-traktori_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_house_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-prikolice-i-poluprikolice_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-kombi-i-laka-dostavna-vozila_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-motorcikli_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-kamioni-preko-7-5-t_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_flat_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-traktori_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_land_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-kombajni_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godina-proizvodnje-svi-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-kamioni-do-7-5-t_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-kamperi-i-kamp-prikolice_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-kombi-i-laka-dostavna-vozila_double_mv &

```

```

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_garage_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_room_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-kamioni-preko-7-5-t_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}priceValue_rsd_double &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-prikolice-i-poluprikolice_long_mv &

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stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-kombajni_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}broj-radnih-sati-kombajni_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-agro-prikolice_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_cottage_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-kamperi-i-kamp-prikolice_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}godiste-kamioni-do-7-5-t_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-cm3-havarisani-automobili_string_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_garage_sale_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-traktori_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-autobusi_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_luxury_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}registrovan-do-kombi-i-laka-dostavna-vozila_long_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-autobusi_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}kubikaza-kamperi-i-kamp-prikolice_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}facility_area_range_business_rent_double_mv &

stats.field={!min%3Dtrue+max%3Dtrue}jacina-motora-traktori_double_mv} hits=5754 status=0 QTime=215

```

Currently there are 3 solr nodes, but only 1 replica:

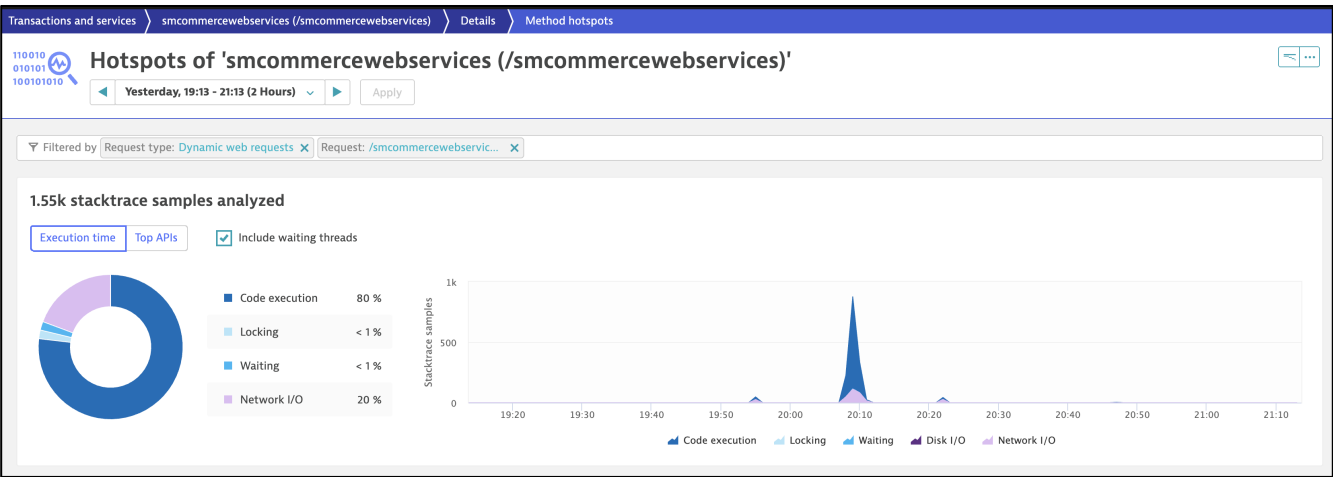
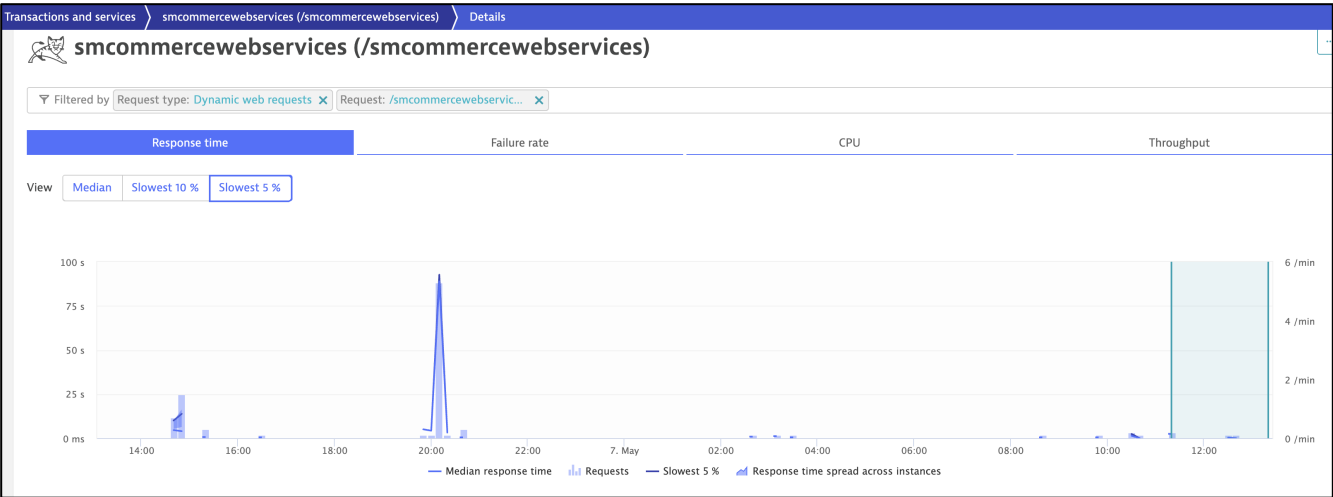
○ master_backoffice_backoffice_product_flip	○ shard1	● solr-1.solr.default.svc.cluster.local (N)
○ master_backoffice_backoffice_product_flop	○ shard1	● solr-0.solr.default.svc.cluster.local (N)
○ master_smrs_smrs_flip	○ shard1	● solr-2.solr.default.svc.cluster.local (N)
○ master_smrs_smrs_flop	○ shard1	● solr-0.solr.default.svc.cluster.local (N)

Recommendations SOLR-01

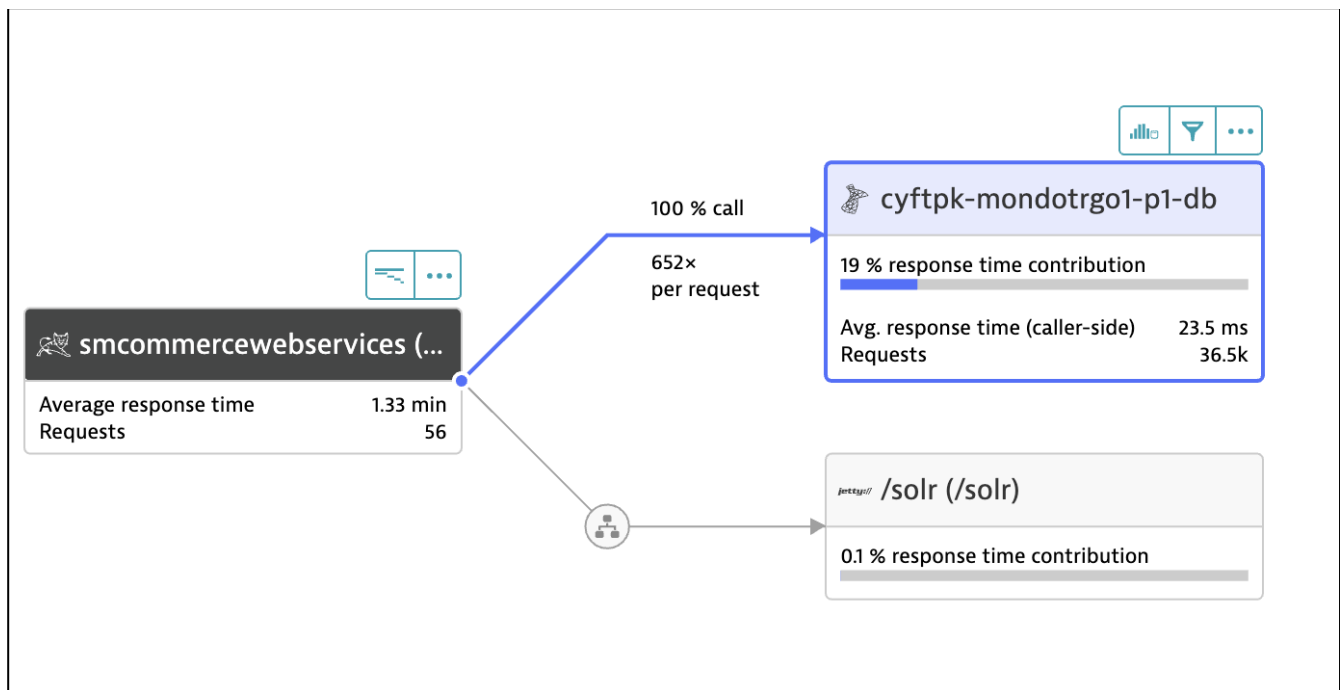
- Increase number of replicas.
- Investigate if all search requests from frontend application really need facet values.
- Try to limit amount of facet attributes if possible from business perspective.
- Limit amount of facets for general search.
- Limit amount of facets for /smcommercewebservices/v2/smrscatalogs/categories
- Conduct end-to-end performance tests with scenario real customer path to properly scale infrastructure

WISHLIST ENDPOINT

Response times and throughput



Service flows shows that there are 652 DB queries per request:



There are three populators which significantly influence response time: ProductStockPopulator, ProductClassificationPopulator, SmCustomCategoryHierarchyPopulator

✓ WishlistController.getWishList	Java rs.sasomange.commercewebservices.v2.controller	58.8 %	...
✓ SmWishlistFacadeImpl.getWishListProductsAndCategories	Java rs.sasomange.facades.wishlist.impl	58.7 %	...
✓ SmWishlistFacadeImpl.populateProducts	Java rs.sasomange.facades.wishlist.impl	38 %	...
✓ DefaultModifiableConfigurablePopulator.populate	Java de.hybris.platform.converters.impl	36.8 %	...
✓ DefaultPopulatorList.populate	Java de.hybris.platform.converters.impl	36.4 %	...
> ProductStockPopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	23.2 %	...
> ProductClassificationPopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	9.99 %	...
> ProductGalleryImagesPopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	1.35 %	...
> ProductCategoriesPopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	1.35 %	...
> ProductPricePopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	0.39 %	...
> ProductReviewsPopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	0.06 %	...
> ProductPriceRangePopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	0.19 %	...
> ProductDeliveryModeAvailabilityPopulator.populate	Java de.hybris.platform.commercefacades.product.converters.populator	0.19 %	...
> AbstractConverter.convert	Java de.hybris.platform.converters.impl	1.23 %	...
> SmWishlistFacadeImpl.populateWishListCategories	Java rs.sasomange.facades.wishlist.impl	20.1 %	...

SmCustomCategoryHierarchyPopulator is doing recursive iteration over categories which can generates high number of queries:

WishlistController\$\$FastClassBySpringCGLIB\$\$e98a6e85.invoke	Java rs.sasomange.commerceservices.v2.controller		58.8 %	...
WishlistController.getWishList	Java rs.sasomange.commerceservices.v2.controller		58.8 %	...
SmWishlistFacadeImpl.getWishListProductsAndCategories	Java rs.sasomange.facades.wishlist.impl		58.7 %	...
SmWishlistFacadeImpl.populateProducts	Java rs.sasomange.facades.wishlist.impl		38 %	...
SmWishlistFacadeImpl.populateWishlistCategories	Java rs.sasomange.facades.wishlist.impl		20.1 %	...
SmWishlistFacadeImpl.populateCategories	Java rs.sasomange.facades.wishlist.impl		20.1 %	...
SmCustomCategoryHierarchyPopulator.populate	Java rs.sasomange.facades.populators		19.8 %	...
SmCustomCategoryHierarchyPopulator.recursive	Java rs.sasomange.facades.populators		19.7 %	...
SmCustomCategoryHierarchyPopulator.recursive	Java rs.sasomange.facades.populators		19.7 %	...
SmCustomCategoryHierarchyPopulator.recursive	Java rs.sasomange.facades.populators		19 %	...
SmCustomCategoryHierarchyPopulator.recursive	Java rs.sasomange.facades.populators		13.7 %	...
SmCustomCategoryHierarchyPopulator.populateCategoryHierarchyData	Java rs.sasomange.facades.populators		12.1 %	...
AbstractSmPromotionSubscribedProductPopulator.populate	Java rs.sasomange.facades.promotionsubscribedproduct.converters.populators		11.3 %	...
AbstractUrlResolver.resolve	Java de.hybris.platform.commerceservices.url.impl		0.52 %	...
CategoryModel.getName	Java de.hybris.platform.category.model		0.26 %	...

Recommendation - WISHLIST

Validate list of populators executed by SmWishlistFacadeImpl.getWishListProductsAndCategories method. Try to remove or refactor populators populatorsProductStockPopulator, ProductClassificationPopulator, SmCustomCategoryHierarchyPopulator

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